

Team 17: Where Healing Happens - Mapping Mental Health in Childhood Cancer

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Project Goal



To explore and visualize the relationship between environmental spatial characteristics surrounding an individual and their mental health outcomes using GIS and spatial analysis.



Mental Health Constructs

- Psychological Stress
 - Anxiety
 - Depression
- Quality of Life
 - Physical Function Mobility
 - Pain Intensity
- Perceived Social Support
 - Overall
 - Significant Other, Family, & Friends



Geographic Information System Information

- Pollution
- Child Opportunity Index
- Neighborhood Tapestry
- Green Spaces
- Misc. Population Demographics

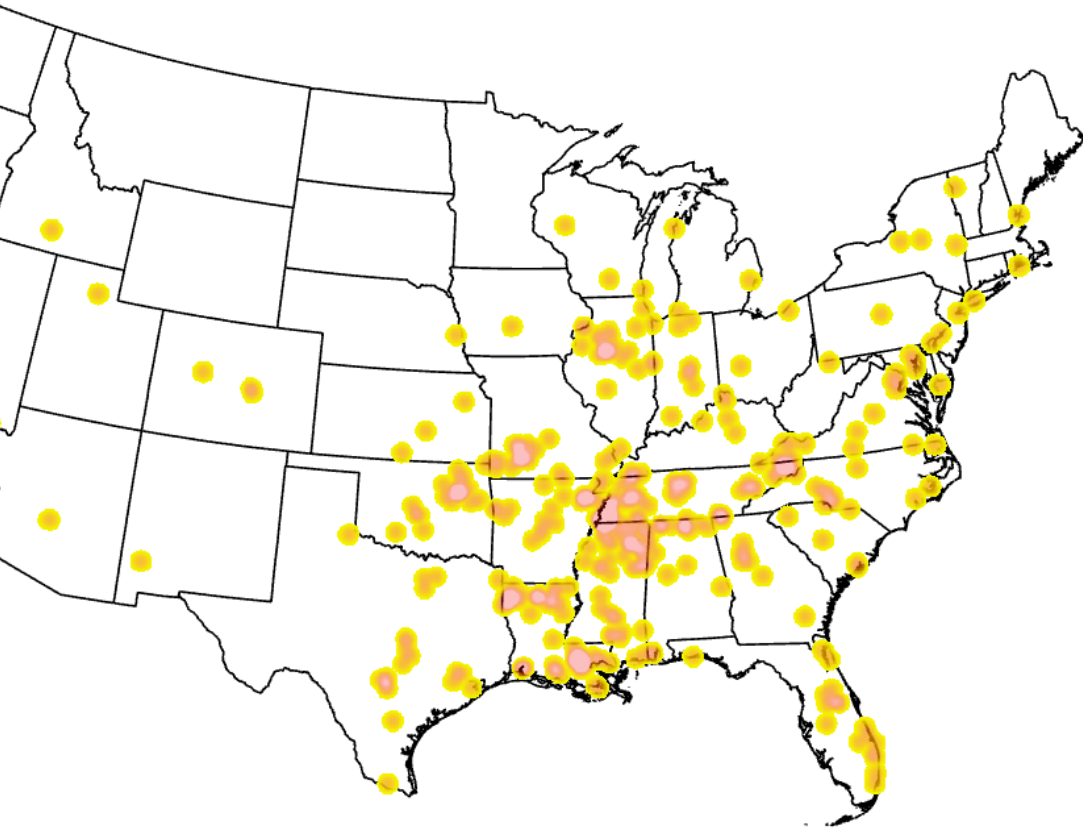


Methods: Mental Health

- Measures collected as part of a larger St Jude initiated protocol
 - Principal Investigator: **Rachel Webster, PhD**
 - Department: Psychology & Biobehavioral Sciences
- PROMIS: Anxiety, Depression, Physical Mobility & Pain Intensity
 - PROMIS® Pediatric Profile General Population v3.0 – Profile-48
 - PROMIS® General Population v2.1 – Profile-43
 - Scored using the HealthMeasures Scoring Service
- MSPSS: Multidimensional Scale of Perceived Social Support
 - 12-item Scale
 - Zimet GD, Dahlem NW, Zimet SG, Farley GK. The Multidimensional Scale of Perceived Social Support. *Journal of Personality Assessment* 1988; 52:30-41.



Methods: GIS



- Geocode de-identified patient locations and display them spatially in GIS.
- Create patient location maps showing the geographic distribution of the study population.
- Overlay patient locations with environmental factors, including NLCD land cover, green space, air pollution (PM_{2.5}, SO₂, NO₂, CO), noise pollution, and AQI.
- Map socioeconomic and neighborhood characteristics, such as the Child Opportunity Index (COI).
- Create buffer zones around patient locations to measure nearby environmental exposures.
- Develop layered maps that allow users to view and compare multiple environmental and social variables.



Sample Descriptives



	13-18 years (N = 240)	19-24 years (N = 250)
Race		
White	163 (67.9)	164 (65.6)
Black or African American	46 (19.2)	47 (18.8)
American Indian or Alaska Native	1 (0.4)	1 (0.4)
Asian	2 (0.8)	7 (2.8)
Other	22 (9.2)	22 (8.8)
Prefer not to say	6 (2.5)	9 (3.6)
Ethnicity		
Not Hispanic, Latino/a, or of Spanish origin	211 (89.4)	221 (89.5)
Hispanic, Latino/a, or of Spanish origin	23 (9.7)	26 (10.5)
Prefer not to say	2 (0.8)	0 (0.0)
Gender		
Male	116 (48.5)	103 (42.0)
Female	117 (49.0)	134 (54.7)
Other	6 (2.4)	8 (3.2)



Sample Descriptives



	13-18 years (N = 240)	19-24 years (N = 250)
Highest Education		
Less than High School	182 (80.2)	6 (2.4)
High School or GED	31 (13.6)	93 (37.6)
Some College or Technical School	5 (2.2)	85 (34.4)
College Graduate	1 (0.4)	62 (25.1)
Graduate Degree	0 (0.0)	1 (0.4)
Employment Status		
Work Full Time	3 (1.3)	105 (42.5)
Work Part Time 15-34 hours	28 (11.9)	48 (19.4)
Work Part Time less than 15 hours	21 (8.9)	15 (6.1)
Do Not Currently Work for Pay	179 (75.8)	73 (29.6)

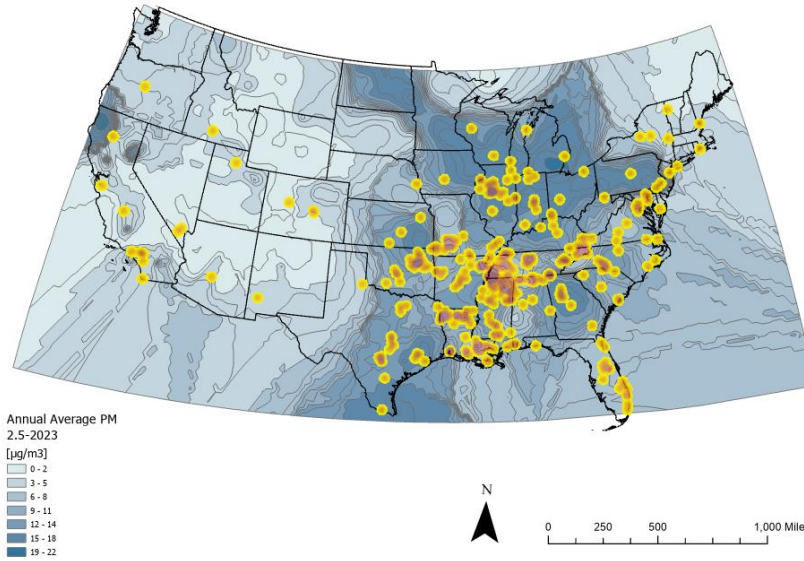


Results – Air Pollution

PM_{2.5}

- All PM_{2.5} are categorized as “good” (0-12) or “moderate” (12.1-35.5)

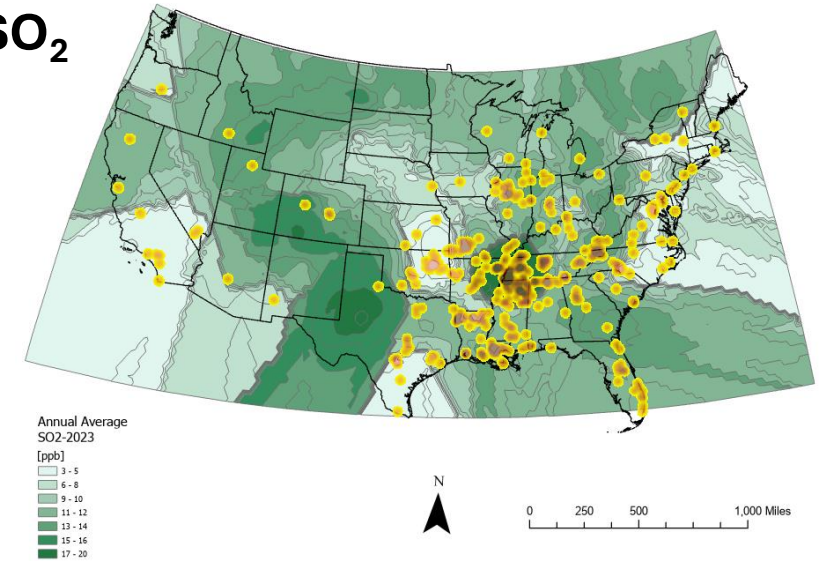
Insufficient
variability



Annual Average SO₂

- All SO₂ are categorized as “good” (0-35 ppb)

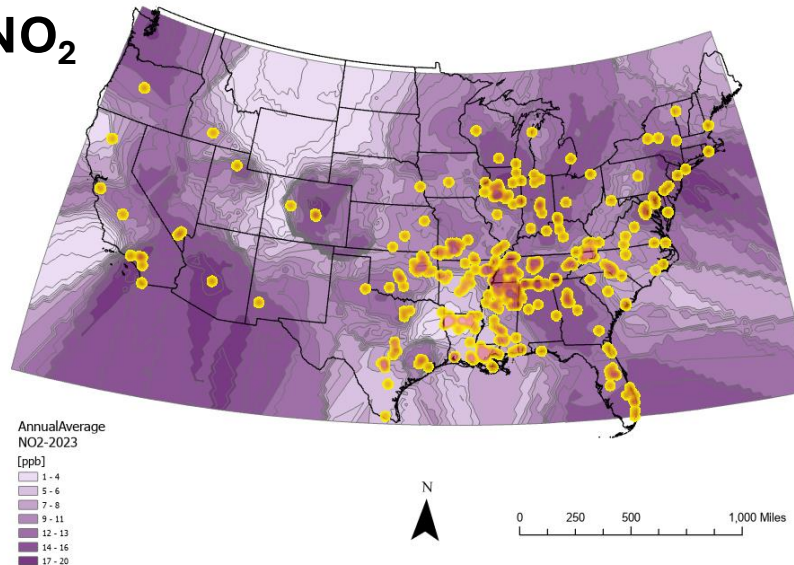
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variability



Annual Average NO₂

- All NO₂ are categorized as “good” (0-53 ppb)

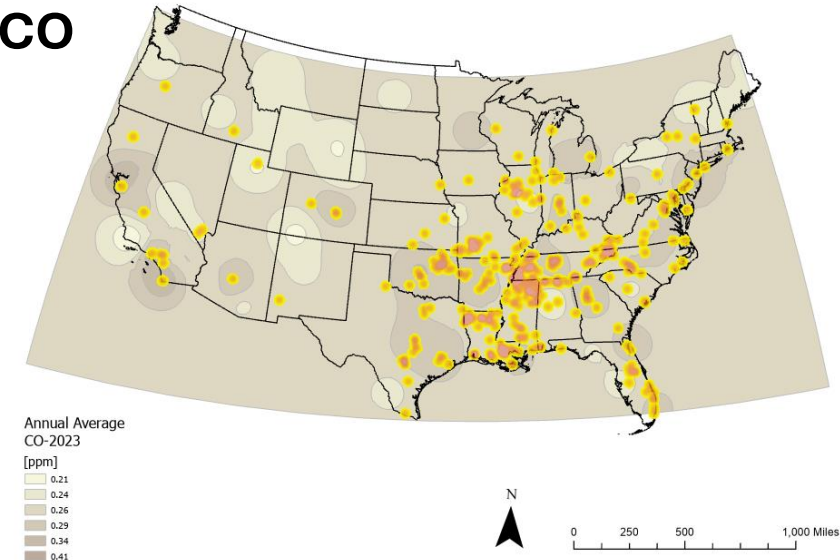
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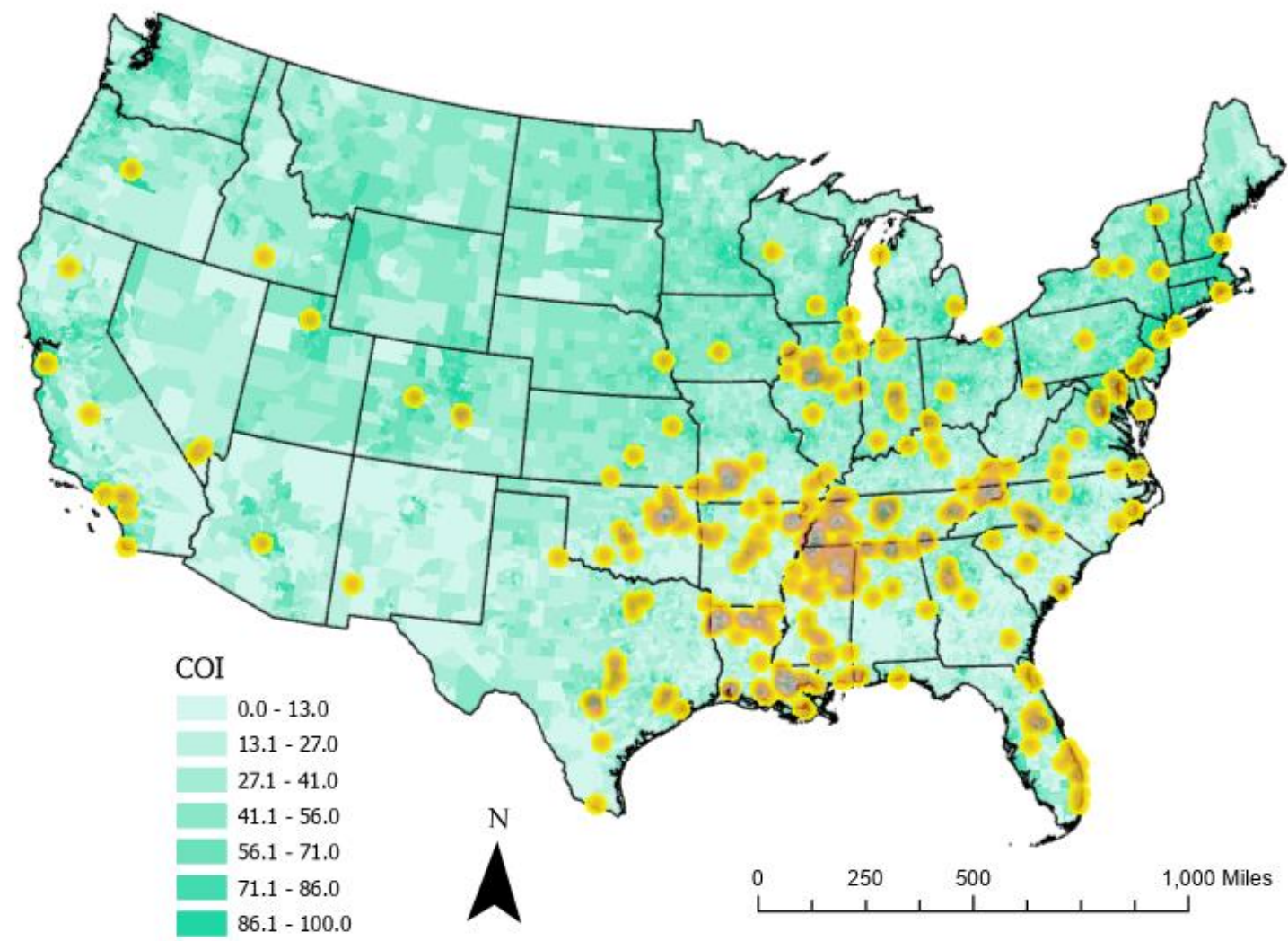
Annual Average CO

- All CO are categorized as “good” (0-4.4 ppm)

Insufficient
variability



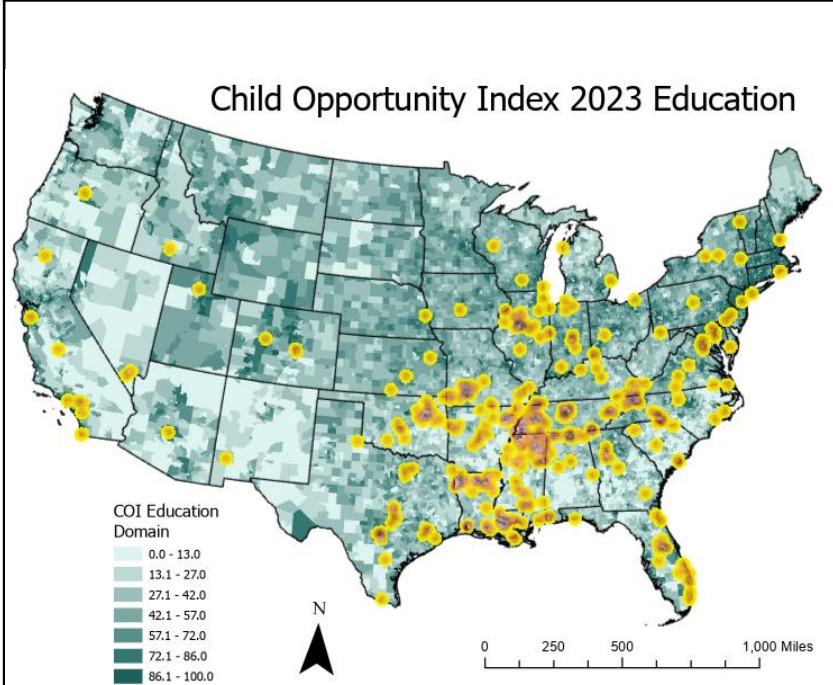
Results – Child Opportunity Index and MSPSS



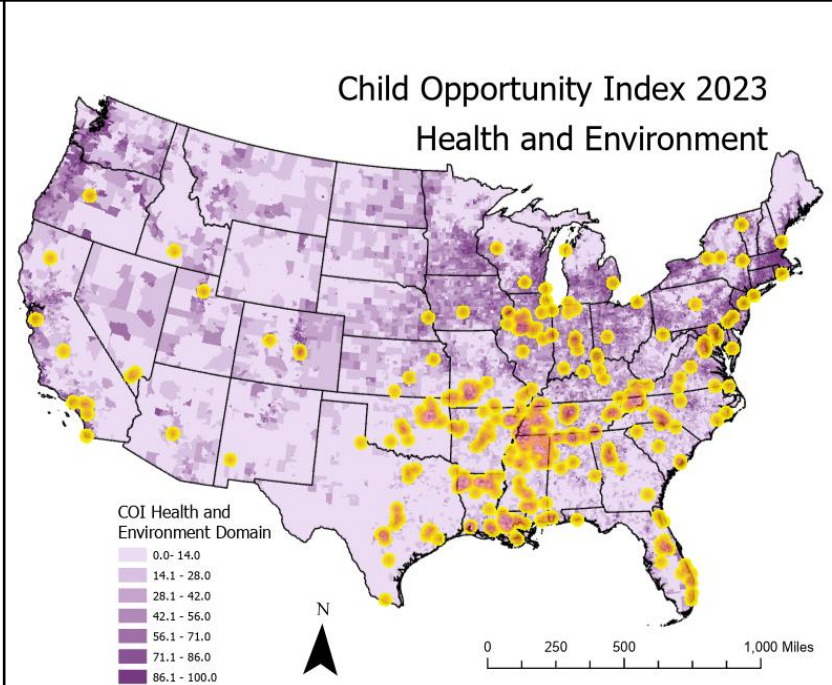
MSPSS	r	p
Total	0.156	.001
Significant Other	0.145	.001
Family	0.140	.002
Friends	0.097	.033



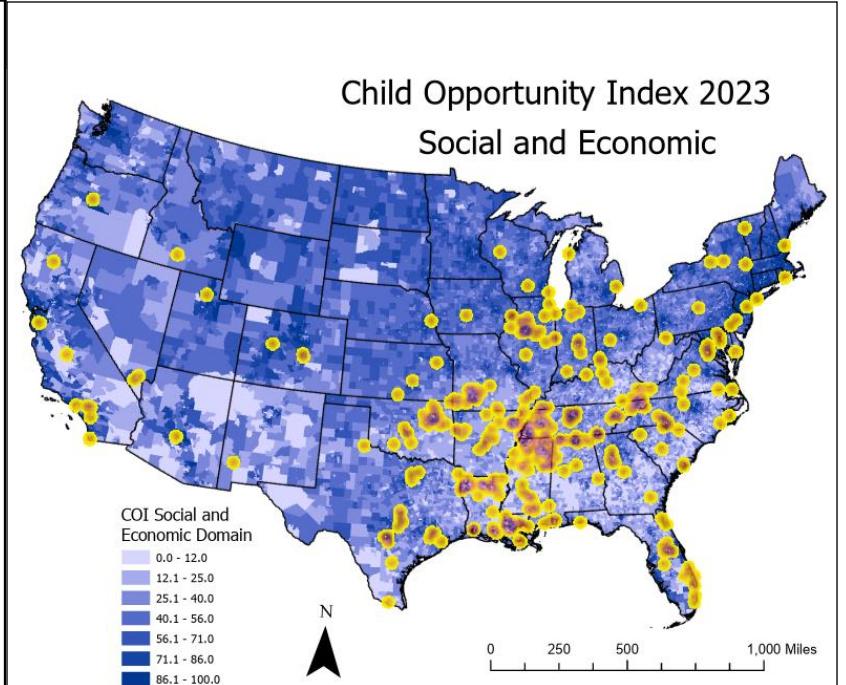
Results – Child Opportunity Index and MSPSS



MSPSS	r	p
Total	0.153	.001
Significant Other	0.138	.002
Family	0.143	.002
Friends	0.192	.043

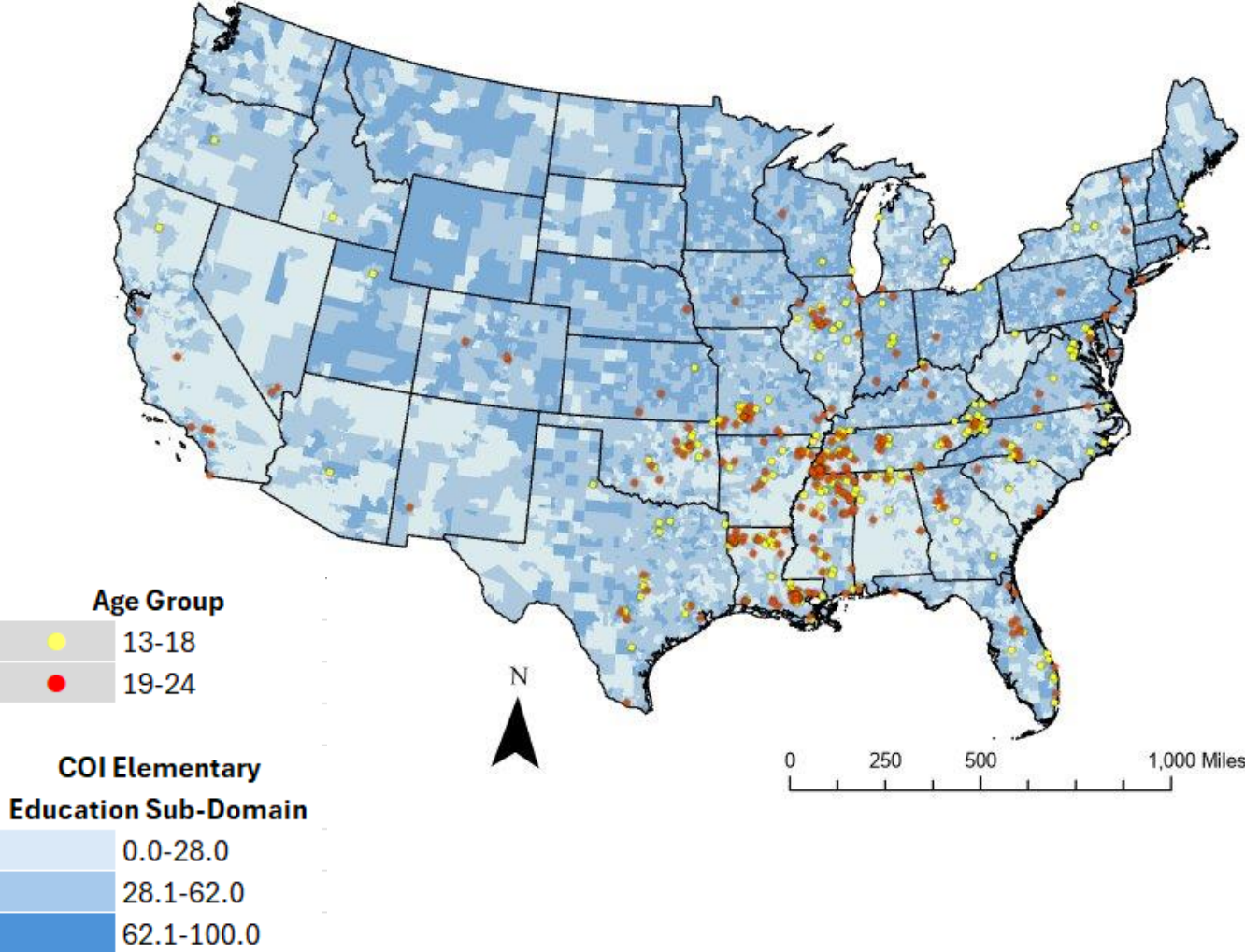


MSPSS	r	p
Total	0.146	.001
Significant Other	0.138	.002
Family	0.118	.009
Friends	0.088	.052



MSPSS	r	p
Total	0.141	.002
Significant Other	0.132	.004
Family	0.122	.007
Friends	0.093	.041

Results – Child Opportunity Index vs Age Group



COI	F- Value	<i>p</i>
COI Social & Economic	4.00	.046
COI Elementary Education	5.17	.023
COI Employment	4.68	.031



Final Thoughts and Future Directions



- Connections to St Jude Values
 - Especially 1, 4, 5, 7
- Future Directions
 - Explore additional:
 - Psychology constructs
 - Spatial data sets
 - Patient populations
 - Global geographical reach





Thank You!



Rachel Webster, PhD